Aaron Wei

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Education

University of British Columbia Sep. 2026 – Present

M.Sc. in Computer Science (Ph.D. Track)

- Supervisor: Dr. Danica J. Sutherland. Research areas: kernel methods, statistical machine learning.

University of British Columbia Sep. 2021 – May 2026

B.Sc. in Mathematics and Computer Science, Honours

- Dean's List 2025, 2026. Graduate-level coursework in mathematics, computer science, and statistics.

Relevant Experience

Research Assistant in Statistical Machine Learning May 2025 – Present

Group of Dr. Danica J. Sutherland, UBC

- Developing machine learning methods and theory for problems in hypothesis testing and causal inference.

Projects

Uncertainty quantification for functional regression | *Python, Gaussian Processes* Jun. 2026 – Present

- Developing algorithms and theoretical foundations for inference on function-valued data (e.g., conditional treatment effects, acoustics), complemented by uncertainty quantification using Bayesian methods.

Kernel two-sample tests on unpaired data | *PyTorch, Hypothesis Testing* Jun. 2025 – May 2026

- Created an objective for learning power-maximizing feature transformations in two-sample testing, resulting in a first-author publication in *Transactions on Machine Learning Research* (link) and a workshop contribution at the *43rd International Conference on Machine Learning*.
- Optimizing a deep neural network on this objective resulted in a 17–23% increase in test power compared to previous benchmarks for determining the homogeneity of image data (CIFAR-10 vs. CIFAR-10.1).
- Extended the large-sample theory of U-statistics (a class of unbiased estimators) to analogous results for unpaired data.

Full-stack campus facilities app (course project) | *TypeScript, Node.js, React* Jan. – Apr. 2026

- Built a RESTful API to handle structured data for 300+ campus facilities and 4,000+ course sections, with unit and end-to-end tests implemented using Mocha and Chai.
- Wrote a React interface for user queries along with visualizations for historical grade trends using Recharts.

Awards

UBC BPOC Award: $1,500 academic scholarship. 2026

UBC Enhanced Summer Research Experience Award: $3,500 for Faculty of Science research. 2025

UBC Advanced Machine Learning Training Network Award: $5,000 for ML research. 2025

Livingston International Scholarship: $1,000 academic scholarship. 2025

Technical Skills

Languages: Python, C, MATLAB, Java, Type/JavaScript

Software/Libraries: NumPy, PyTorch, Scikit-Learn, Pandas, OpenCV, Google Cloud Platform

Other Experience

Reviewer May 2026

43rd International Conference on Machine Learning, Workshop on Hypothesis Testing

- Provided feedback on paper submissions related to modern hypothesis testing techniques.

Instructor Jun. – Nov. 2024

Hillhouse Education

- Assisted K-12 students in learning high school subjects and basic programming in Python, in turn increasing their conceptual understanding and exam performance.

Software Team Member Sep. 2022 – Dec. 2023

UBC Uncrewed Aircraft Systems

- Wrote Python programs with OpenCV and Pytesseract for image processing and optical character recognition.